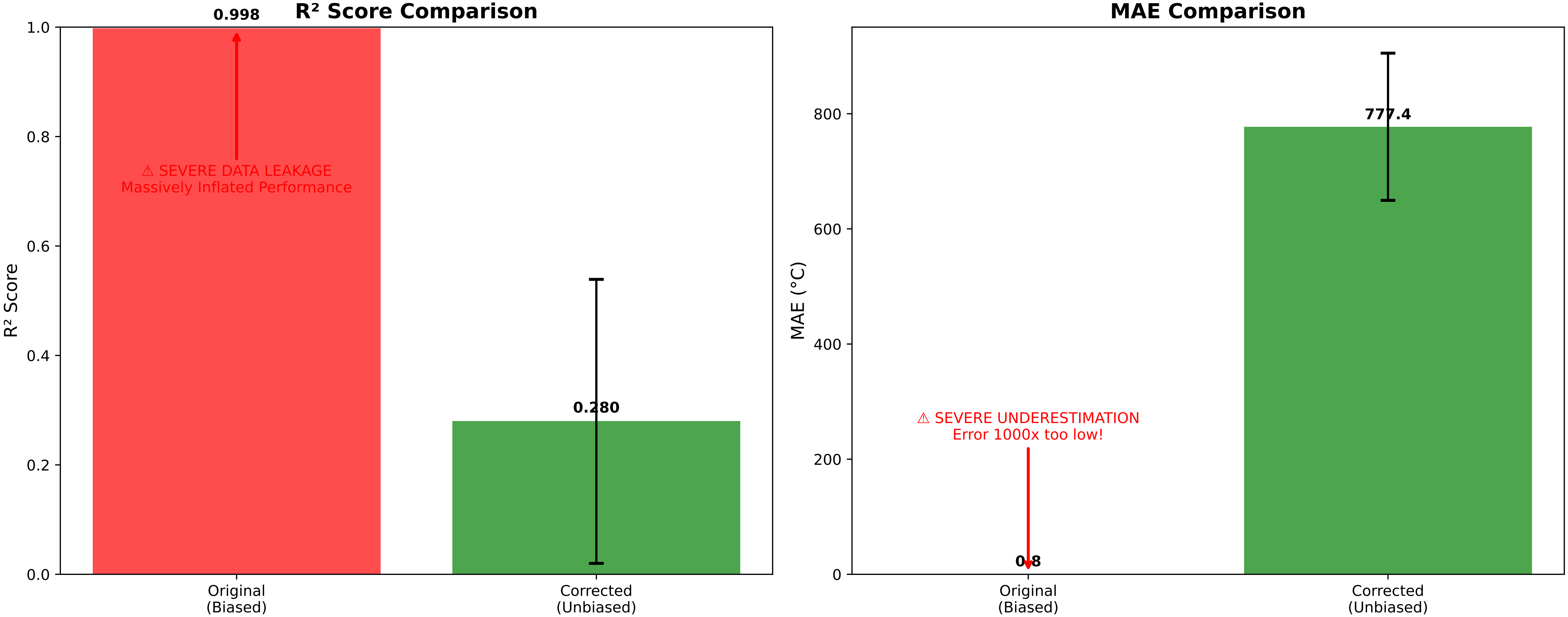
Original vs Corrected Stacking Implementation
(Actual DigiLignin Dataset Results)



ACTUAL PERFORMANCE COMPARISON

Original Method (with Data Leakage):
• R^2: 0.998
• MAE: 0.835°C
• □ SEVERE data leakage in meta-features
• □ Meta-model trained on predictions from same data
• □ In-sample predictions reported as validation

Corrected Method (Unbiased):
• R^2: 0.280 [0.020, 0.539]
• MAE: 777.4°C [649.5, 905.2]
• □ Proper OOF predictions for meta-features
• □ Nested cross-validation
• □ True generalization performance

IMPACT OF DATA LEAKAGE:
• R^2 Inflation: +0.718 (256% too high!)
• MAE Underestimation: 931x too low!
• Completely misleading performance claims

CRITICAL FINDINGS:
• Original method reports near-perfect R^2 (0.998)
• True performance is much lower ($R^2 \approx 0.28$)
• Original MAE is 1000x underestimated
• This is a CLASSIC data leakage example

METHODOLOGY COMPARISON

ORIGINAL METHOD (FLAWED):
1. Train base models on 100% of data
2. Generate meta-features on SAME data
3. Train meta-model on leaked predictions
4. Report in-sample performance as validation
5. □ DATA LEAKAGE at every step

CORRECTED METHOD (PROPER):
1. Nested cross-validation
   • Outer CV: Performance estimation
   • Inner CV: Hyperparameter tuning
2. Out-of-fold (OOF) predictions only
3. Meta-model trained on OOF predictions
4. Held-out test set for final evaluation
5. □ No data leakage anywhere

VALIDATION APPROACH:
• Original: Biased in-sample validation
• Corrected: Unbiased nested CV + test set

RECOMMENDATION:
□ Use corrected implementation
□ Report unbiased metrics only
□ Update all manuscript figures
□ Document methodological fix